⇒ Inverse functions:

ex:- $g(x) = 3x-2$.

→ step 1, convert $g(x)$ to y.

$$y = 3x-2$$

step2 → In place of y write the x.

$$x = 3y-2$$

Now make y seprate / subects.

$$\frac{x+2}{3} = y$$ → so now this y is actual inverse

so $$\boxed{g^{-1}(x) = \frac{x+2}{3}}$$

⇒ Now let's prove this using graphically.

$$y = 3x-2$$

if $x=0$

$y = 3(0)-2$

$\boxed{y=-2}$

if $x=1$ then

$y = 3(1)-2$

$\boxed{y=1}$

x	y
0	-2
1	1
2	4

Now find x and y intercepts:

→ y intercepts then $x=0$.

then $y=-2$ ✓

x intercepts then $y=0$.

$0 = 3x-2$

$2 = 3x$

$= \boxed{\frac{2}{3} = x}$ 0.666